



EDUCATION

- 2025 **MIT-WHOI Joint Program in Oceanography, Ph.D.**
Biological Oceanography
Thesis: A Lagrangian Perspective of Mesoscale Biophysical Interactions in the Subtropical Ocean (<https://hdl.handle.net/1721.1/158812>)
Committee: Drs. Michael J. Follows (Advisor; MIT), Stephanie Dutkiewicz (MIT), Amala Mahadevan (WHOI), Colleen Mouw (URI)
- 2018 **Temple University, B.A.** (*cum laude*, top 9% in College of Science and Technology)
University Honors Program; Major: Mathematics; Minors: Physics, Business
Honors Scholar Thesis: Impact of Warm Water Anomalies on Phytoplankton Composition in the Santa Barbara Channel (dx.doi.org/10.34944/dspace/4610)
Committee: Drs. Raphael Kudela (Advisor; UCSC), Henry Houskeeper (Advisor; UCSC, WHOI), Eric Cordes (TU)

RESEARCH POSITIONS

University of Hawai'i at Mānoa, SOEST, Department of Oceanography

- 2025 – POSTDOCTORAL RESEARCHER, Powell Lab
- Using the MOM6-COBALT Hawai'i Island Chain Regional Model to study the vertical transport of nitrate by anticyclonic eddies and the associated phytoplankton response
 - Collaborating with NOAA Pacific Island Fisheries scientists to examine how ocean dynamics around the Hawaiian Islands support local fisheries

Massachusetts Institute of Technology, Department of Earth, Atmospheric & Planetary Sciences

- 2024 – 2025 POSTDOCTORAL ASSOCIATE, Follows Marine Biogeochemical Modeling Group
- Researched the role of mesoscale physical features in shaping phytoplankton abundances using amplicon sequence data and Lagrangian trajectory simulations
 - Generated an idealized 1/10° simulation of the North Pacific Ocean with MITgcm-Darwin
- 2019 – 2024 GRADUATE RESEARCH ASSISTANT, Follows Marine Biogeochemical Modeling Group
- Sampled North Pacific microbial communities on the SCOPE-Gradients 4 cruise and examined how fine-scale ocean transport shapes community composition
 - Studied how the structure of mesoscale eddies affects phytoplankton concentrations, differentiating features that trap from those actively mixing with satellite remote sensing

University of Southern California, Department of Biological Sciences

- 2022 VISITING GRADUATE RESEARCH ASSISTANT, Fuhrman Marine Microbial Ecology Group
- Developed protocols for genomic internal standards, performed DNA extractions, and ran PCR for SCOPE-Gradients 4 cruise samples

Woods Hole Oceanographic Institution, Biology Department

- 2019 VISITING GRADUATE RESEARCH ASSISTANT, Sosik Optical Ocean & Phytoplankton Ecology Lab
- Validated satellite phytoplankton function type algorithms with flow cytometry imagery
 - Processed and extracted chlorophyll samples from SPIROPA research cruises

Temple University, Department of Biology

2018 – 2019 POST-GRADUATE RESEARCH ASSISTANT, Kulathinal Evolutionary Genomics & Biological Informatics Lab

- Studied the evolution of duplicate (“de novo”) genes in the *Drosophila* lineage and their functions
- Updated the open-source website PlayBIGData.org in HTML, including increasing functionality and developing educational modules on big data and coding fundamentals

2015 – 2018 UNDERGRADUATE RESEARCH ASSISTANT, Kulathinal Evolutionary Genomics & Informatics Lab

- Studied genetic variation and the intensity of natural selection in *Drosophila melanogaster* with a population genomics approach

NASA Jet Propulsion Laboratory, Earth Science Division

2018 PROJECT LEAD, NASA DEVELOP National Program

- Served as lead of an early-career research project, which included providing coding and data analysis assistance, and writing weekly progress reports to NASA administrators
- Evaluated environmental factors driving habitat range shifts for the California grunion with satellite remote sensing observations and in situ citizen science data
- Developed a predictive model of the potential size of grunion spawning events and GUI interface application for local government and academic end-users

NASA Armstrong Flight Research Center & University of California, Irvine

2017 INTERN, NASA Student Airborne Research Program

- Quantified changes in phytoplankton community composition and biomass as a response to decadal ocean temperature shifts and environmental fluctuations in the Santa Barbara Channel with MATLAB and satellite remote sensing observations
- Assisted in the operation of instruments onboard a NASA Airborne Science Program aircraft to sample atmospheric chemicals and image land and water surfaces

PUBLICATIONS

Peer-Reviewed

6. **Jones-Kellett AE**, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (2026). The dynamic mesoscale sink and source niches for eukaryotic phytoplankton in a subtropical gyre. *PNAS*. 123(25): e2608700123. <https://doi.org/10.1073/pnas.2608700123>
 - [Stirring the Gyre: How ocean mixing sustains hidden diversity](#), CBIOMES News
5. Graff van Crevelde S, Coesel SN, Lavoie E, Iverson V, Morales R, Schatz M, **Jones-Kellett AE**, McNichol JC, Key R, Fuhrman JA, Durham BP, Armbrust EV (2026). Induced pathogenicity toward open-ocean diatoms by a filterable bacterium *Ekbidna algicida* sp. nov. *The ISME Journal*. 20(1): wrag038. <https://doi.org/10.1093/ismejo/wrag038>
4. **Jones-Kellett AE** & Follows MJ (2025). The satellite chlorophyll signature of Lagrangian eddy trapping varies regionally and seasonally within a subtropical gyre. ♦ *Ocean Science*, 21, 1141-1166. <https://doi.org/10.5194/os-21-1141-2025>
 - ♦ Highlight paper, as selected by editors
 - [Satellite Eyes on Swirling Seas](#), CBIOMES News
3. Koestner D, Clayton S, Lerner P, **Jones-Kellett AE**, Walker SL (2025). Biogeochemical-Argo floats reveal seasonality of the biological carbon pump influenced by the Lofoten Basin Eddy. *Geophysical Research Letters*, 52(12): e2024GL111937. <https://doi.org/10.1029/2024GL111937>
 - [How does a persistent eddy impact the biological carbon pump?](#), OCB Highlight
2. **Jones-Kellett AE**, McNichol JC, Raut Y, Cain KR, Ribalet F, Armbrust EV, Follows MJ, Fuhrman JA (2024). Amplicon sequencing with internal standards yields accurate picocyanobacteria cell abundances

as validated with flow cytometry. *ISME Communications*, 4(1): ycae115.

<https://doi.org/10.1093/ismeco/ycae115>

- [Innovative Sequencing Technique Enhances Marine Microbial Research](#), CBIOMES News

1. **Jones-Kellett AE** & Follows MJ (2024). A Lagrangian Coherent Eddy Atlas for Biogeochemical Applications in the North Pacific Subtropical Gyre. *Earth System Science Data*, 16, 1475-1501. <https://doi.org/10.5194/essd-16-1475-2024>

In Preparation

2. McGinty N, Hu Y, **Jones-Kellett AE**, McNichol JC, Merrow K, Finkel Z, Irwin A. Biogeography and stoichiometry of mesozooplankton in the (sub)tropical Pacific Ocean.
1. **Jones-Kellett AE**, McNichol JC, Raut Y, Berube PM, Fuhrman JA, Follows MJ. Meridional Stirring of Temperature-Adapted Prochlorococcus Ecotypes. *In prep for Limnology & Oceanography Letters*.

TEACHING & MENTORING

Teaching

Case Western Reserve University, Dept. of Earth, Environmental, & Planetary Sciences

Apr 22, 2026 GUEST LECTURER, Introduction to Oceanography

Apr 26, 2023

Massachusetts Institute of Technology, Dept. of Earth, Atmospheric, & Planetary Sciences

2022 TEACHING ASSISTANT, Mechanisms and Models of the Global Carbon Cycle

Other

2019 – GUEST SCIENCE SPEAKER, 20+ K-12th grade classes

2017 – 2019 MATHEMATICS TUTOR, High school level Algebra I, Algebra II, & Geometry

Research Advisees

University of Hawai'i at Mānoa

FISHERIES GRADUATE PROGRAM

2026 – MeiLin Precourt (co-advised with Drs. Justin Suca & Brian Powell)

Massachusetts Institute of Technology

UNDERGRADUATE RESEARCH OPPORTUNITIES PROGRAM

2024 Khyatee Atolia (Wellesley College), “Which Satellite Products Best Capture Ocean Currents in the North Pacific?”

2021 Sydney Kim (MIT), “How do Eddies Modify the Ocean’s Uptake of CO₂?”

EAPS MASTER OF SCIENCE PROGRAM

2022 – 2023 Christine Padalino (co-advised with Drs. Mick Follows & Greg Britten), “The Effect of Eddies on fCO₂ in the North Pacific Surface Ocean”:

<https://hdl.handle.net/1721.1/159266>

Career Mentees

MIT EAPS Peer Mentor Program

2023 – 2024 Lucy Brock (Undergraduate Student)

2021 – 2023 Lucy Sandoe (Masters Student)

2021 – 2022 Kelly McKeon (Ph.D. Student)

MIT EAPS Application Mentorship Program & MIT-WHOI Joint Program Application Support

2019 – 2021 Prospective Ph.D. Students (anonymized for privacy)

HONORS & AWARDS

2025	NSF Postdoctoral Research Fellowship, Recommended for Award (declined)
2025	Ocean Science Highlight Paper Designation (https://doi.org/10.5194/os-21-1141-2025)
2024	MIT EAPS Community Builder Award
2023	Temple University 30 Under 30
2022	FilaChange Student Travel Grant (~\$1,000)
2019	MIT EAPS John H. Carlson Fellowship (~\$84,000)
2019	NSF Graduate Research Fellowship Program, Honorable Mention
2018	Ocean Sciences Meeting Student Travel Grant (\$500)
2017	Temple University Diamond Award
2017	Temple University Creative Arts Research and Scholarship Grant (\$4,000)
2016, 2017	David Tepper, CST '64 and Elaine Kowalewski Scholarship in Mathematics (\$3,000)
2016	Temple University Merit Scholarship Educational Enhancement Stipend (\$4,000)
2016	White Haven Lions Club Scholarship (\$500)
2014	Temple University Merit Scholarship (partial tuition, 4 years)

OCEANOGRAPHIC CRUISES

Nov 18 – Dec 15, 2021	R/V Thomas G. Thompson, SCOPE-Gradients 4, North Pacific
Sep 20 – 22, 2019	R/V Neil Armstrong, MIT-WHOI Joint Program Cruise, Northeast Atlantic
Jun 28 – Jul 7, 2019	SS/V Corwith Cramer, MIT-WHOI Joint Program Cruise, Northeast Atlantic
Jun 28 – 29, 2017	Fishing Vessel, NASA SARP Cruise, Santa Barbara Channel

INVITED TALKS

4. **University of Washington School of Oceanography**, Banse Seminar. (Feb 26, 2025). The Phytoplankton Response to Lagrangian Eddy Trapping in the North Pacific Subtropical Gyre.
3. **BIOcean5D Collaboration**, Lagrangian Modelling Workshop. (Sep 09, 2024). Lagrangian water mass advection histories explain phytoplankton variability at the mesoscale.
2. **NASA Goddard Ocean Ecology Lab**, MPOWIR NASA Speaker Series. (Jun 24, 2024). Lagrangian Interpretations of Mesoscale Biophysical Interactions in the North Pacific Subtropical Gyre.
1. **Woods Hole Oceanographic Institution**, Mahadevan Group Meeting. (Oct 10, 2023). Lagrangian Eddy Trapping Fosters Chlorophyll Hot Spots in the North Pacific Subtropical Gyre.

CONFERENCE PRESENTATIONS (first author only; +equal contribution)

24. Jones-Kellett AE, Matthews D, Friedrich T, Powell B (Jun 15, 2026). Physical Mechanisms Driving Phytoplankton Blooms in Anticyclonic Hawaiian Lee Eddies. Poster. *CBIOMES Annual Meeting*. New York, NY, USA.
23. Jones-Kellett AE, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (Jun 15, 2026). Phytoplankton absolute gene abundances reflect mesoscale ocean dynamics. Oral. *CBIOMES Annual Meeting*. New York, NY, USA.
22. Jones-Kellett AE, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (Oct 03, 2025). Eukaryotic phytoplankton are sustained by eddies and lateral mixing in the open ocean. Oral. *OceanParcels 10-Year Anniversary Meeting*. Utrecht, NL.
21. Jones-Kellett AE, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (Jun 17, 2025). Revealing mesoscale organization of communities with metabarcoding on the Gradients 4 transect. Oral & Poster. *CBIOMES Annual Meeting*. New York, NY, USA.

20. Jones-Kellett AE, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (Mar 30, 2025). Subtropical Phytoplankton Community Composition Reflects Lateral Mixing History and Mesoscale Eddies. Oral. **ASLO Aquatic Sciences Meeting**. Charlotte, NC, USA.
19. Jones-Kellett AE, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (Nov 12, 2024). NPSG phytoplankton community composition from G4 reflects lateral mixing history and mesoscale eddies. Oral & Poster. **SCOPE Annual Meeting**. New York, NY, USA.
18. Jones-Kellett AE, McNichol JC, Raut Y, Fuhrman JA, Follows MJ (Feb 21, 2024). The Fluid Dynamical Structuring of Microbial Communities Along an Eastern North Pacific Transect. Poster. **Ocean Sciences Meeting**. New Orleans, LA, USA.
17. Jones-Kellett AE & Follows MJ (Nov 10, 2023). The Lagrangian History of the Biologically Anomalous Cyclone Cathy (Station 4) from the Gradients 4 Cruise. Poster. **SCOPE Annual Meeting**. New York, NY, USA.
16. Jones AE & Follows MJ (Jun 5, 2023). Enhanced Biological Activity in Lagrangian Coherent Eddies of the North Pacific Subtropical Gyre. Oral. **ASLO Aquatic Sciences Meeting**. Palma de Mallorca, Spain.
15. Jones AE & Follows MJ (Aug 31, 2022). Satellite Chlorophyll Signatures of Eddy Coherency in the North Pacific Subtropical Gyre. Oral. **FilaChange**. Providence, RI, USA.
14. Jones AE & Follows MJ (Mar 4, 2022). Does coherency shape the chlorophyll signature of North Pacific subtropical gyre eddies? Oral. **Ocean Sciences Meeting**. Virtual.
13. Jones AE & Follows MJ (Jan 25, 2022). Does eddy coherency affect plankton populations in the NPSG? Oral. **SCOPE Annual Meeting**. Virtual.
12. Jones A +, Knapp H +, Peacock A +, Wakamatsu L +, Holt B (Dec 10, 2018). Predicting Grunion Migration Patterns and Spawning Areas in Response to Changes in California's Oceans by Coupling Satellite and In Situ Data. Poster. **American Geophysical Union Fall Meeting**. Washington, DC, USA.
11. Jones A +, Knapp H +, Peacock A +, Wakamatsu L +, Holt B (Aug 1, 2018). Predicting Grunion Migration Patterns and Spawning Areas in Response to Changes in California's Oceans. Poster. **NASA Annual Earth Science Application Showcase**. Washington, DC, USA.
10. Jones AE, Ranz JM, Kulathinal RJ (Apr 12 & 14, 2018). Evolution of de novo genes in the *Drosophila melanogaster* lineage. Poster. **59th Annual Drosophila Research Conference**. Philadelphia, PA, USA.
9. Jones AE, Ranz JM, Kulathinal RJ (Apr 12, 2018). Evolution of de novo genes in the *Drosophila melanogaster* lineage. Poster. **Temple Undergraduate Research Forum and Creative Works Symposium**. Philadelphia, PA, USA.
8. Jones AE, Houskeeper HF, Kudela RM (Feb 15, 2018). Impact on phytoplankton composition in the Santa Barbara Channel from the 2013-2015 warm water anomaly. Poster & lightning talk. **Ocean Sciences Meeting**. Portland, OR, USA.
7. Jones AE, Ranz JM, Kulathinal RJ (Jan 20, 2018). Evolution of de novo genes in the *Drosophila melanogaster* lineage. Poster. **Harvard National Collegiate Research Conference**. Cambridge, MA, USA.
6. Jones AE, Houskeeper HF, Kudela RM (Oct 6, 2017). Warm water anomaly effect on Santa Barbara Channel phytoplankton composition. Poster. **College of Science and Technology 8th Annual Temple Undergraduate Research Symposium**. Philadelphia, PA, USA.
5. Jones AE, Houskeeper HF, Kudela RM (Aug 8, 2017). Effect of changing sea surface temperature on phytoplankton composition in the Santa Barbara Channel. Oral. **NASA Student Airborne Research Program Meeting**. Irvine, CA, USA.
4. Jones AE, Stanley CE, Kulathinal RJ (Apr 20, 2017). A dynamic and adaptive male genomic landscape in *Drosophila*. Oral. **Temple Undergraduate Research Forum and Creative Works Symposium**. Philadelphia, PA, USA.
3. Jones AE, Stanley CE, Kulathinal RJ (Apr 1, 2017). Functional genomic landscape in *Drosophila* provides evidence for pervasive adaptation of sexually selected male traits. Oral. **93rd Annual Meeting of the Pennsylvania Academy of Science**. Wilkes Barre, PA, USA.
2. Jones AE, Chin JL, Stanley CE, Kulathinal RJ (Sep 16, 2016). Adaptive functional landscape of reproductive genes in *Drosophila* provides evidence for positive selection on sperm-specific proteins.

Poster. *College of Science and Technology 7th Annual Temple Undergraduate Research Symposium*. Philadelphia, PA, USA.

1. Jones AE, JL Chin, NH Rigby, CE Stanley, RJ Kulathinal (Jul 29, 2016). Functional landscape of locally adaptive reproductive proteins in *Drosophila melanogaster*. Poster. *Temple University Biology Department Summer Undergraduate Research Program Poster Session*. Philadelphia, PA, USA.

WORKSHOP PARTICIPATION

2026	Simons CBIOMES Workshop on Machine Learning for Time Series Data, NY, NY
2025	OceanParcels 10-year Anniversary Event, Utrecht, NL
2024	NASA/OCB PACE Hackweek, University of Maryland Baltimore County, Baltimore, MD
	Simons CBIOMES Workshop on Numerical Circulation & Ecosystem Modeling, NY, NY
2023	GO-BGC/BGC-Argo Float Data Workshop, UMass Boston, Boston, MA
	Simons CBIOMES Workshop on Transects & Eco-Provinces, MIT, Cambridge, MA
2021	TIDE Seminar: Racism, Colonialism, & Extraction within the Geosciences
	Unlearning Racism in Geoscience
2020	IOCCG 2020 Summer Lecture Series (Virtual Adaptation)

INTERNAL PRESENTATIONS

Feb 02, 2026	UH Mānoa Department of Oceanography, Oceanography Seminar
Aug 13, 2025	MIT EAPS, Afternoon Talk Series
Aug 08, 2023	
Jul 14, 2025	Simon's Collaboration on Ocean Processes and Ecology, Gradients series
Mar 11, 2024	
Oct 02, 2023	
Mar 12, 2025	Simon's Collaboration on Computational Biogeochemical Modeling of Marine Ecosystems
Nov 15, 2023	
Sep 14, 2022	
Feb 13, 2025	MIT EAPS, Student Seminar
Oct 19, 2023	
Apr 14, 2023	
Oct 07, 2022	
Oct 26, 2021	
Nov 16, 2023	WHOI, Biology Dept. Seminar
Apr 29, 2021	

PROFESSIONAL SERVICE

<u>Reviewer for:</u>	Nature Communications, Geophysical Research Letters, , Ocean Science, Journal of Geophysical Research: Oceans, Bioscience, PLOS One
2025	ASLO 2025 Aquatic Sciences Meeting Session Chair: "Microbial Responses to Pulse Disturbances in Aquatic Environments"
2023 – 2024	MIT EAPS Toward Inclusion and Diversity (TIDE), Co-Organizer
2023	MIT EAPS-SCC Faculty Search, Graduate Student Advisory Group Member
2020 – 2022	MIT EAPS Let's Invest in Neighborhood K-12 (LINK-12), Co-Founder
2020 – 2022	MIT EAPS Student Advisory Committee, Public Service Chair
2021	EAPS Application Mentorship Program, Outreach Coordinator
2020 – 2021	MIT EAPS Diversity, Equity, and Inclusion Committee, Member

PROFESSIONAL MEMBERSHIP

2025 –	MPOWIR Mentor Group
2019 – 2025	MIT EAPS Toward Inclusion and Diversity (TIDE)
2019 – 2025	MIT Women in Course XII
2016 – 2018	Temple University Mathematics Club
2016 – 2018	Temple University Association for Women in Mathematics

PUBLIC OUTREACH

K-12 Volunteer Scientist Speaker (UH Outreach, MIT EAPS LINK-12, Skype-a-Scientist, etc.)

Apr 01, 2026	Ali'iolani Elementary Career Day, Honolulu, HI, USA
Nov 22, 2024	Stephanie Magistro 9 th -12 th Grade, Charlestown High School, MA, USA
Dec 13, 2023	Jessica Crane 5 th Grade, Kelly Elementary School, MA, USA
May 17, 2023	Elizabeth Jones Kindergarten, Rice Elementary, PA, USA
May 20, 2022	
Feb 18, 2021	
May 19, 2020	
Jan 20, 2023	Christine Nicholson 5 th Grade, Chenery Middle School, MA, USA
Oct 22, 2020	STEM Story Time 1 st -3 rd Grade, LET'S GO Boys and Girls, Inc, DC, USA
Oct 16, 2020	Dolores Simmons 4 th Grade, Saint Mary's School, Vancouver, BC
Oct 15, 2020	Jessica Lincecum Kindergarten, Green Elementary, OH, USA
Sep 18, 2020	Caitlin Ward 12 th Grade, Berkshire School, MA, USA
Jun 02, 2020	Briana Button Kindergarten, Rice Elementary, PA, USA
Jun 01, 2020	Michelle Brooks-Rogers Kindergarten, Rice Elementary, PA, USA
May 28, 2020	Yvonne Barley Kindergarten, Rice Elementary, PA, USA
May 21, 2020	Jennifer Detweiler Kindergarten, Rice Elementary, PA, USA
May 20, 2020	Nicole Sivilli 4 th Grade, Harmony School, NJ, USA
May 18, 2020	
Apr 20, 2020	Aaron Huber 7 th Grade, Cayman International, Cayman Islands
Nov 02, 2019	Girls Day at the MIT Museum, Cambridge, MA

John H. Carlson Lecture Science Exhibit Volunteer

Oct 24, 2024	New England Aquarium Simons Theatre, Boston, MA, USA
Oct 19, 2023	
Nov 14, 2019	

MEDIA COVERAGE

2026	Stirring the Gyre: How ocean mixing sustains hidden diversity , CBIOMES News
2025	How does a persistent eddy impact the biological carbon pump? , Ocean Carbon & Biogeochemistry Highlight
	Satellite Eyes on Swirling Seas , CBIOMES News
2024	Innovative Sequencing Technique Enhances Marine Microbial Research , CBIOMES News
2023	Research and Technology Innovator: Alexandra Jones , Temple University 30 Under 30
2023	How This Grad Student Shifted Her Student Loan Strategy through the Pandemic , Personal Finance for PhDs Podcast
2019	Hey Beacher, Leave Those Fish Alone , Hakai Magazine
2018	Surfing with the Silversides* , YouTube
	*2025 Update: Video has been removed by NASA